

EX.NO: 8

IMPLEMENTATION OF K-NN CLASSIFICATION ALGORITHM

AIM:

To classify the test data to a target attribute based on the model built using the training data with the help K-NN classification algorithm.

PROCEDURE:

1. Install the necessary package in R Studio such as “class”
2. Set the current working directory
3. Save the dataset in the current working directory
4. Read the dataset in the form of csv file using **read.csv ()**
5. Finally create the model using kNN algorithm with the help of training data.

SOURCE CODE:

```
# Loading data
data(iris)

# Structure str(iris)

# Installing Packages
Install.Packages("e1071")
install.packages("caTools")
install.packages("class")

# Loading package
library(e1071)
library(caTools)
library(class)
```

```
# Loading data

data(iris)

head(iris)

# Splitting data into train and test data

split <- sample.split(iris, SplitRatio = 0.7)

train_cl <- subset(iris, split == "TRUE")

test_cl <- subset(iris, split == "FALSE")

# Feature Scaling

train_scale <- scale(train_cl[, 1:4])

test_scale <- scale(test_cl[, 1:4])

head(train_scale)

head(test_scale)

# Fitting KNN Model to training dataset

classifier_knn <- knn(train = train_scale,
                      test = test_scale,
                      cl = train_cl$Species, k = 1)

classifier_knn
```

OUTPUT:

```
File Edit Code View Plots Session Build Debug Profile Tools Help
RStudio
Source
Environment History Connections Tutorial
Data
iris 150 obs. of 5 variables
test_c1 60 obs. of 5 variables
train_c1 90 obs. of 5 variables
train_scale 90 obs. of 5 variables
values
classifier_jmnn
split

# Loading package
library(ggplot2)
library(caret)
library(class)

# Loading data
data(iris)
head(iris)
# Splitting data into train and test data
split <- sample.split(iris, splitRatio = 0.7)
train_c1 <- subset(iris, split == "TRUE")
test_c1 <- subset(iris, split == "FALSE")

# Feature scaling
train_scale <- scale(train_c1[, 1:4])
test_scale <- scale(test_c1[, 1:4])

# Head(train_scale)
# Head(test_scale)

# Fitting klm model to training dataset
classifier_jmnn <- klm(train = train_scale,
  test = test_scale,
  k = 1)

# Predictions
predict(classifier_jmnn, test_scale)
```

RESULT:

Thus, the test data are classified successfully based on the model built using K-NN classification algorithm.